AMINOPHYLLINE

Check if on oral theophylline

Slow IV — arrhythmias

CHEMICAL NAME Theophylline–ethylenediamine (2:1) complex
CLASS Methylxanthine bronchodilator (IV form of theophylline)
TRADE NAMES Phyllocontin · Aminophylline injection · Truphylline

THERAPEUTIC USES / INDICATIONS

- Acute severe or life-threatening asthma not responding to nebulised bronchodilators and steroids (senior decision).
- Acute exacerbation of COPD (rarely).
- Apnoea of prematurity (historical).

ADMINISTRATION

- **IV loading dose** over at least 20 min (rapid → arrhythmias, seizures, death).
- Then continuous infusion by pump.
- Loading dose omitted if patient already on oral theophylline — check level first.
- Use ideal body weight in obesity.

DOSAGES

- **Loading:** 5 mg/kg IV over 20 min (max 500 mg).
- **Maintenance:** 0.5 mg/kg/h (reduce to 0.3 in elderly, HF, liver disease, interacting drugs).
- **Children:** 5 mg/kg load, then 1 mg/kg/h (age-dependent).
- Check level 4–6 h after start; target 10–20 mg/L.

PHARMACOKINETICS

- Converted to theophylline (~80% theophylline by weight).
- Metabolised by CYP1A2.
- **t½:** ~8 h (variable).

PHARMACODYNAMICS

- Phosphodiesterase inhibition and adenosine antagonism → bronchodilation, respiratory stimulation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting.
- Tachycardia, palpitations.
- Headache, restlessness.

ADVERSE EFFECTS (SERIOUS)

- Arrhythmias, hypotension.
- Seizures.
- Hypokalaemia (with β2 agonists).
- Death with rapid injection or toxicity.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to theophylline or ethylenediamine.
- Acute porphyria.
- **Caution:** arrhythmias, epilepsy, hyperthyroidism, liver disease.

KEY INTERACTIONS

- Same as theophylline: ciprofloxacin, macrolides, cimetidine ↑ levels; smoking, rifampicin ↓ levels.

NURSING MANAGEMENT

- Ask about oral theophylline before loading dose.
- Continuous ECG monitoring; BP, HR, RR, SpO_2.
- Check theophylline level at 4–6 h and daily.
- Monitor K^+ (with salbutamol) every 12–24 h.
- Watch for vomiting, agitation, seizures.

PATIENT EDUCATION

- Report palpitations, nausea or feeling shaky during the drip.

KEY POINTS

- IV theophylline for life-threatening asthma.
- Load 5 mg/kg over 20 min — skip if on theophylline.
- Maintenance 0.5 mg/kg/h; ECG monitoring.
- Watch levels and K^+.

SOURCES CHECKED **[460]** Aminophylline 25 mg/ml solution for injection/infusion (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158) **[461]** Aminophylline, IV (US label) **[462]** Aminophylline, IV (US label, Hospira)
Full references p. 432–457